MILITARY SPECIFICATION

MAINTENANCE KIT, ELECTRONIC EQUIPMENT MK-578/APX

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers Maintenance Kit, Electronic Equipment MK-578/APX, which is designed for shop use in servicing Video Decoder KY-364/APX. The equipment covered by this specification performs the function normally prescribed for a test bench harness when testing and servicing the KY-364/APX.

1.2 Classification - The Maintenance Kit, Electronic Equipment MK-578/APX is of one type and shall consist of the following items:

Items	Reference Designation	Appl. Para.
Board, extension (Line Amplifier)	E-1	3.4.1
Board, extension (Gate)	E-2	3.4.1
Board, extension (Line Driver)	E-3	3.4.1
Board, extension (Code Select)	E-6/7	3.4.1
Board, extension (Bracket and Passive)	E-8	3.4.1
Board, extension (Garble)	E-9	3.4.1
Cable Assembly	W-1	3.4.2
Case Assembly	(See Note 6.4)	3.4.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-H-15362	Harnesses; Test Bench, Avionic Equipment, General Specification for

FSC 5841

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MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200, unless otherwise specified herein.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and material shall be obtained as outlined in Specification MIL-T-21200, except as otherwise specified below:

(1) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of Specification MIL-T-21200 are not met.

(2) For nonstandard parts of an electrical nature, as defined in Specification MIL-T-21200, an itemized list of the parts with source and, where applicable, contractor's drawing or specification shall be submitted in lieu of other specification requirements.

(3) Approval to use nonstandard parts or material in the Video Decoder KY-364/APX shall be regarded as approval to use the same item in the Maintenance Kit MK-578/APX, and shall not necessitate re-submittal to the Bureau of Naval Weapons for approval.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction, and workmanship, except as otherwise specified herein.

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3.2.2 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 15 pounds.

3.2.3 Operating Life

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for an indefinite number of hours, continuously or intermittently.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200, except as otherwise specified herein.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.6 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ±10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ±0.5 VDC

3.2.8 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-H-15362.

3.2.9 Voltage Protection - No provisions for voltage protection to the equipment shall be provided in the test harness assembly.

3.3 Performance - The maintenance kit shall provide the interconnecting cabling and extension boards required to bench test a Video Decoder KY-364/APX. The cable assembly shall be used to connect a 28-volt DC power source to the KY-364/APX, and to provide leads for radar control signals and the decoder front panel lighting circuit. The extension boards shall serve to position the plug-in subassemblies of the KY-364/APX, except for delay line subassemblies, in a manner to facilitate trouble-shooting and testing procedures.

3.4 Detail Requirements

3.4.1 Boards, Extension E-1, E-2, E-3, E-6/7, E-8 and E-9 - The extension boards shall meet the following requirements:

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3.4.1.1 Function - The extension boards shall be suitable for elevating plug-in circuit subassemblies of the KY-364/APX to an accessible position for trouble-shooting and testing purposes. Use of the extension boards shall in no way alter the electrical performance of the KY-364/APX equipment being tested. One board shall be supplied for each type of plug-in circuit subassembly employed in the KY-364/APX. Extension boards for delay line subassemblies are not required. Extension boards in the maintenance kit shall consist of laminated plastic circuit boards having etched copper conductors, etched copper plug-in contactors on one end, and a multipin female connector on the other end. The etched copper conductors on the board shall consist only of leads from pins of the plug-in contactors to corresponding pins of the female connector. Plug-in contactors and connectors on the extension boards shall be polarized in the same manner as the connectors of the corresponding plug-in subassembly of the KY-364/APX with which each board is used.

3.4.1.2 Form Factor - The extension board assemblies shall be constructed of a nominal 1/16-inch thick laminated plastic. Each extension board shall be a minimum of 5 inches long, and shall not exceed the width of the corresponding plug-in subassembly board of the KY-364/APX. The thickness of the plug-in contactor end of the board shall be such that no damage will result when the contactors are inserted into the receptacle on the chassis of the KY-364/APX.

3.4.1.3 Weight - The total weight of all extension boards shall not exceed three pounds.

3.4.1.4 Extension Board Conductors - Conductors and plug-in contactors shall be etched on one or both sides of the extension boards in copper. Accommodation of KY-364/APX circuits and the characteristics of conducted signals shall determine whether one or both sides of the board are etched.

3.4.1.5 Extension Board Connectors - Connectors on extension board assemblies shall duplicate connectors on corresponding plug-in subassemblies of the KY-364/APX.

3.4.1.6 Extension Board Marking - A reference designation shall be assigned to each extension board, in order to relate the board to the corresponding subassembly and circuit connector of the KY-364/APX. The numerical portion of the reference designation for each board shall coincide with the numerical portion of the A and J designation for the related KY-364/APX subassembly and connector. Extension board assemblies shall be marked as follows:

Extension Board	Reference Designation
a. Line Amplifier	E-1
b. Active Read Gate	E-2
c. Line Driver	E-3
d. Code Select	E-6/7
e. Bracket and Passive	E-8
f. Garble	E-9

3.4.2 Cable Assembly W-1 - The cable assembly shall meet the following requirements:

3.4.2.1 Function - The cable shall be used to connect a 28-volt DC power source to the KY-364/APX and to provide leads for radar control signals and the decoder front panel lighting circuit.

3.4.2.2 Form Factor - The cable assembly shall be 6 feet in length.

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3.4.2.3 Weight - The weight of the cable assembly shall not exceed 5 pounds.

3.4.2.4 Cable Assembly Characteristics - The cable assembly shall consist of eight conductors, a 26-pin connector on one end to mate with receptable J17 on the KY-364/APX, and insulated spade lugs on the other end of all leads. Wire used in the cable assembly shall have the characteristics and be of a size required for satisfactory operation of the KY-364/APX. Strain relief for the cable shall be provided. Lead-to-pin connections for the cable assembly are given in Table I.

TABLE I

Cable Assembly Lead-To-Pin Connections

Lead No.	Function	Pin
SX132A20	28-volt DC	F
SX131A20	DC Ground	G
SX136A20	Mode 1	H
SX137A20	Mode 2	J
SX138A20	Mode 3	K
SX139A20	Ground	L
Light	Panel Lighting	W
SX130A20	Ground	X

3.4.2.5 Cable Assembly Marking - The cable assembly shall be marked for identification in accordance with Specification MIL-N-18307, except as otherwise specified herein. The multipin connector end and each lead at the other end of the cable assembly shall be marked by approved means, including the use of pressure-sensitive tape, appropriately marked and protected by clear plastic sleeving, and preprinted preformed plastic bands. Markings shall be of a permanent nature, i.e., not easily removable and not subject to undue deterioriation during normal use.

3.4.3 Case Assembly (*) - The case assembly shall meet the following requirements:

3.4.3.1 Function - The case assembly shall accommodate and store the cable and extension board assemblies of the MK-578/APX.

3.4.3.2 Form Factor - The maximum outside dimensions of the case assembly shall be 6 inches high by 6 inches wide by 9 inches long.

3.4.3.3 Weight - The weight of the case assembly shall not exceed 7 pounds.

3.4.3.4 Case Assembly Characteristics - The case shall provide drip-proof protection when the case lid has been placed in position and clamped. The case shall be provided with quick-opening latches that satisfy the requirements of Specification MIL-T-21200. The main position of the latch shall be on the case body; the smaller portion (hook) shall be on the case lid. The case lid shall be hinged to the case body in a manner as to be permanently attached. Handles on the case assembly shall not be required. The case assembly shall be fabricated of a lightweight metal to provide suitable protection and strength for normal storage in laboratory or maintenance areas.

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3.5 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

Video Decoder KY-364/APX

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating, and their applied circuit voltage, current, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high-temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted, and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Test - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, modifying the test limits of the applicable tests to correspond with the limits specified herein.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall be responsible for accomplishing acceptance tests. The following acceptance tests shall be conducted in accordance with test procedures approved by the Bureau of Naval Weapons:

(1) Individual Tests

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4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4 Test Procedures - The procedures used for conducting preproduction and acceptance tests shall be prepared by the contractor and submitted to the Bureau of Naval Weapons for review and approval. The Bureau of Naval Weapons reserves the right to modify the tests or to require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. Preproduction and acceptance test procedures previously approved by the Bureau of Naval Weapons for the maintenance kit shall be acceptable for use without re-submission for approval.

4.5 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor unitl it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.6 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Use - The electronic maintenance kit covered by this specification is intended for use with standard electronic work benches in testing, adjusting, and repairing the Video Decoder KY-364/APX.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Any deviations from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified herein, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.4.1 It should be noted that the original procurement of the MK-578/APX was not procured in accordance with Specification MIL-N-18307. Any procurement made under this specification shall conform to Specification MIL-N-18307 for all items. Original Extension Boards and case assembly were delivered with the following part numbers:

Extension Board No.	Original Part No.
E-1 Line Amplifier	P-AC 358
E-2 Active Read Gate	P-AC 410
E-3 Line Driver	P-AC 357
E-6/7 Code Select	P-AC 360
E-8 Bracket and Passive	P-AC 359
E-9 Garble	P-AC 361
Cable Assembly (W-1)	P-AC 362
Case Assembly (*)	P-34D5

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specification or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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